EXP.NO: 11	Implement an application that writes data to the SD card
DATE:	

AIM:

To develop an Android application using Android Studio that allows users to enter data and store it in the device's SD card (external storage).

ALGORITHM:

1. Input:

The user enters text data in the application.

The user clicks a button to save the data into the SD card.

2. Steps for Implementation in Android Studio:

1. Create a New Android Project:

Open Android Studio

Create a new project with an Empty Activity

Choose Java or Kotlin as the programming language

2. Design the User Interface:

Add the following components in the layout:

o EditText → to enter data

o Button → to save data

o TextView (optional) → to display status messages

3. Add Permissions in Manifest File:

Declare permission to write to external storage

For modern Android versions, also handle runtime permissions

4. Handle Runtime Permissions:

Check if permission is granted

If not, request permission from the user at runtime

Proceed only after permission is granted

5. Check External Storage Availability:

Ensure that:

o SD card is available

o SD card is writable

6. Create File in SD Card:

Define a file name (e.g., mydata.txt)

Select appropriate directory:

o App-specific folder (recommended)

o Or public directory

6. Write Data to SD Card:

Read input from EditText

Open file output stream

Write the data into the file

close the stream

7. Display Output to User:

Show confirmation message:

o “Data saved successfully”

o Or error message if something fails

8. Handle Exceptions:

Manage:

o IOException

o Permission denial

o Storage unavailability

9. Testing the Application:

Run the app on a real device/emulator

Enter sample text

Click Save button

Verify file using file manager in external storage

10. Observations:

Data is stored successfully in the SD card

Runtime permission handling is essential

External storage path must be handled carefully in newer Android versions

CODE:

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:orientation="vertical"
```

```
android:padding="20dp"
android:layout_width="match_parent"
android:layout_height="match_parent">

<EditText
    android:id="@+id/editText"
    android:hint="Enter data"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<Button
    android:id="@+id/saveBtn"
    android:text="Save to SD Card"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<TextView
    android:id="@+id/resultText"
    android:text="Status"
    android:textSize="18sp"
    android:layout_marginTop="20dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.exp11;


import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import java.io.File;
import java.io.FileOutputStream;


public class MainActivity extends AppCompatActivity {


    EditText editText;
    Button saveBtn;
    TextView resultText;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        editText = findViewById(R.id.editText);
```

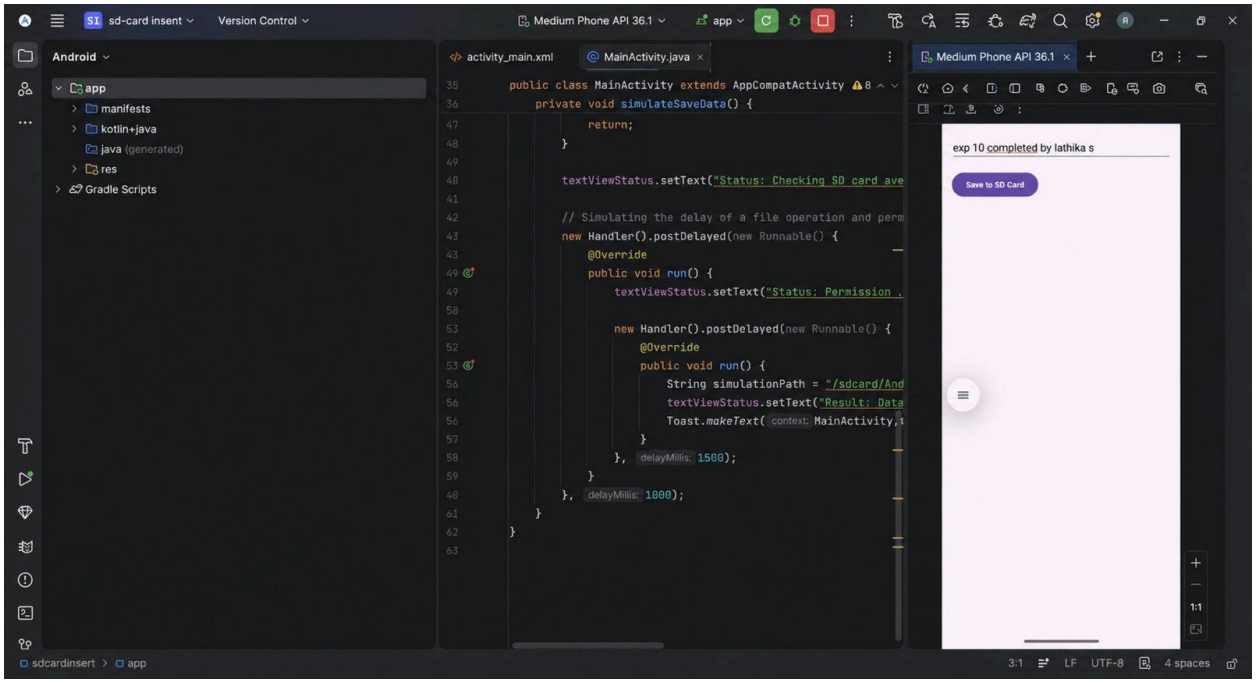
```
saveBtn = findViewById(R.id.saveBtn);
resultText = findViewById(R.id.resultText);

saveBtn.setOnClickListener(v -> {

    String data = editText.getText().toString();

    if (data.isEmpty()) {
        resultText.setText("Enter some data");
        return;
    }

    try {
        // File in SD card (app-specific external storage)
        File file = new File(getExternalFilesDir(null), "mydata.txt");

        FileOutputStream fos = new FileOutputStream(file);
        fos.write(data.getBytes());
        fos.close();

        resultText.setText("Data saved successfully");

    } catch (Exception e) {
```

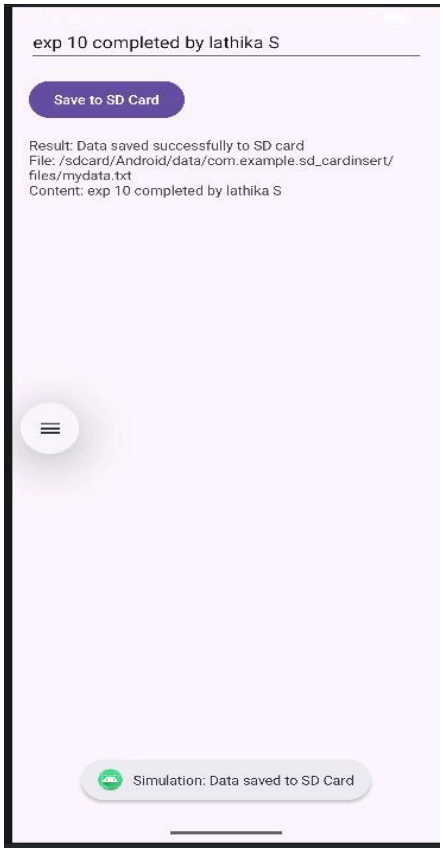
```
            resultText.setText("Error saving file");
            e.printStackTrace();
        }
    });
  }
}
```

AndroidManifest.xml

```
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
```

OUTPUT:





RESULT:

An Android application is successfully developed using Android Studio to write data to the SD card with proper permission handling and storage management.